

Speed & Distance, Races & Tracks

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Speed, Time & Distance

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INTRODUCTION

Speed of an object is defined as the distance travelled by it in per unit time interval.

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

UNITS

PHYSICAL QUANTITY

UNITS

TIME	Hour (hr)	Seconds (sec)
SPEED	km/hr	m/sec
DISTANCE	Kilometers (kms)	Meter (m)

CONVERSION

To convert

1. km/hr to m/sec $\Rightarrow$ multiply by $5/18$

$$\frac{\text{Km}}{\text{Hr}} = \frac{1000 \text{ m}}{60 \times 60 \text{ sec}} = \frac{5 \text{ m}}{18 \text{ sec}}$$

2. m/sec to km/hr $\Rightarrow$ multiply by $18/5$

AVERAGE SPEED

It is defined as the total distance travelled in total time.

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

NOTE

- ☐ When the distance is constant: **Average speed = $2xy/x+y$**
- ☐ When the time taken is constant: **Average speed = $(x + y)/2$;**

RELATIVE SPEED

Speed of one object with respect to another.

Same Direction: When objects are moving in same direction, relative speed is equal to the difference between speeds of two objects, which means that the speed appears to be less.

$$\text{Relative Speed} = | S_1 - S_2 |$$

Opposite Direction: When objects are moving in opposite direction facing each other, relative speed is equal to the addition of the speeds of two objects, which means that the speed appears to be more.

$$\text{Relative Speed} = S_1 + S_2$$


PROBLEM ON TRAINS

Case 1: Train crossing a stationary unit object (pole, standing human being etc.)

$$\text{Time Taken} = \frac{\text{Length of Train}}{\text{Speed of Train}}$$

Case 2: Train crossing a stationary object with length (platform, railway tunnel etc.)

$$\text{Time Taken} = \frac{\text{Sum of Length of Train and object}}{\text{Speed of Train}}$$

Case 3: Train crossing a moving object without length

$$\text{Time Taken} = \frac{\text{Length of Train}}{\text{Relative Speed of Train and moving object}}$$

Case 4: Train crossing a moving object with length.

$$\text{Time Taken} = \frac{\text{Sum of Length of Train and object}}{\text{Relative Speed of Train and moving object}}$$

BOAT & STREAM

Let Person's speed in still water be x km/hr and the speed of current be y km/hr

Speed of the Boat in upstream = $(x - y)$ km / hr

Speed of the boat in downstream = $(x + y)$ km / hr

RACES

It is a competition of speeds.

All the participants in a race are required to run a specific distance; whoever does it in minimum time will be the winner of the race.

When all the participants reach the finishing point at the same time, the race is said to be in a “DEAD HEAT”.

Various Phrases:

1. A beats B by ‘y’ meters – means A reaches the finishing point, B is y meters behind. A wins the race.
2. A beats B by ‘t’ minutes – B takes ‘t’ minutes more as compared to A to finish the race. A wins the race.
3. A can give B a start of ‘y’ meters – A starts from the starting point and B starts ‘y’ meters ahead. Still, if they reach at the same instant, it’s a DEAD HEAT.
4. A can give B a start of ‘t’ minutes – A starts ‘t’ minutes after B starts. Still, if they reach at the same instant, it’s a DEAD HEAT.

Time, Speed & Distance

1. A rabbit takes 22 leaps for every 17 leaps of cat and 22 leaps of a rabbit are equal to 17 leaps of the cat. What is the ratio of the speeds of rabbit and cat?

A. 1 : 1 B. 484 : 289 C. 17 : 22 D. none of these

Let the distance traveled by rabbit in 1 leap =x km.

Let the distance traveled by cat in 1 leap =y km.

ATQ, $22x=17y$

$$\frac{x}{y} = \frac{17}{22}$$

Also, time remains same for both to travel the distance

$$S_1 = \frac{22x}{t} \qquad S_2 = \frac{17y}{t}$$

$$\frac{S_1}{S_2} = \frac{\frac{22x}{t}}{\frac{17y}{t}} \quad \rightarrow \frac{S_1}{S_2} = \frac{22x}{17y}$$

$$\rightarrow \frac{S_1}{S_2} = \frac{1}{1}$$

2. Michel travelled from New York to New Jersey covering total distance of 250 miles in 8 hr. partly by car at 30 miles/hour and rest by train at 35 miles/hour. The distance travelled by car is:

A. 150 miles

B. 80 miles

C. 220 miles

D. 180 miles

Let distance travelled by car is= x km.

So distance travelled by train is 250-x

ATQ,

$$\frac{x}{30} + \frac{250 - x}{35} = 8$$

$$\frac{7x + 6 \times 250 - 6x}{210} = 8$$

$$x + 1500 = 1680$$

$$x = 180 \text{ miles}$$

Formula

$$\text{Time} = \frac{\text{Distance}}{\text{speed}}$$

3. The distance of the college and the home of Rajeev is 80 Km. One day he was late by 1 hour than the normal time to leave for the college, so he increased his speed by 4 km/hr and thus he reached to college at the normal time. What is the changed speed of Rajeev?
A. 28 km/h B. 30 km/h C. 40 km/h D. 20 km/h

Let the original speed of Rajeev = x kmph

Now he increase his original speed by 4

So his increased speed is $=(x+4)$ kmph

ATQ

$$\frac{80}{x} = \frac{80}{x+4} + 1$$

$$\frac{80}{x} = \frac{80 + x + 4}{x + 4}$$

$$80x + 320 = 84x + x^2$$

$$x^2 + 4x - 320 = 0$$

$$x^2 + 20x - 16x - 320 = 0$$
$$(x - 16)(x + 20)$$

$$(x = 16) \text{ or } (x = -20)$$

Hence his increased speed is $16+4=20$ kmph

4. Kriplani goes to school at 20 km/h and reaches the school 4 minutes late. Next time, she goes at 25 km/h and reaches the school 2 minutes earlier than the scheduled time. What is the distance of her school?

A. 10 km

B. 15 km

C. 7 km

D. None

Let the Distance between home and School= d .

Suppose he reaches the School on Time, the Time taken = x minutes

Case 1: When he reaches School 4 minutes late,

Time taken = $x+4$

Case 2: when he reaches office 2 minutes early,

Time taken = $x-2$

Ratio of Speed in both cases = $20:25 = 4:5$

Ratio of Time in both cases = $5:4$

$$(x+4)/(x-2)=5/4$$

$$x=26 \text{ min}$$

Taking case 1

$d = \text{Speed} \times \text{Time}$

$$d = \frac{20 \times (26+4)}{60}$$

$d = 10 \text{ km.}$

Answer (A) 10 km

5. A person X starts from Lucknow and another person Y starts from Kanpur to meet each other. Speed of X is 25 km/h, while speed of Y is 35 km/h. If the distance between Lucknow and Kanpur be 120 km and both X and Y start their journey at the same time, when will they meet?

A. 1 hr later

B. 2 h later

C. $\frac{1}{2}$ h later

D. 3 h later

Lucknow

120 km

Kanpur

Relative speed in opposite direction = $25+35=60$ kmph

So they will meet after $\frac{120}{60} = 2 \text{ hours later}$

6. A Pushpak express leaves Lucknow at 6 am and two hours later another train Bhopal express leaves Lucknow. Both trains arrive Bhopal at 4 pm on the same day. If the difference between their speeds be 10 km/h, What is the average speeds of both the trains over entire route:

- A. 40 km/h B. $44 \frac{4}{9}$ km/h C. $42 \frac{3}{5}$ km/h D. none of these**
- 

7. A dog starts chasing to a cat 2 hours later. It takes 2 hours to dog to catch the cat. If the speed of the dog is 30 km/h, what is the speed of the cat?

A. 10 km/h B. 15 km/h C. 20 km/h D. can't be determined

8. A man covers a certain distance by his own car. Had he moved 6 km/h faster he would have taken 4 hours less. If he had moved 4 km/h slower, he would have taken 4 hours more. The distance is:

A. 240 km

B. 640 km

C. 480 km

D. None of these



9. A train covers a certain distance moving at a speed of 60 km/h. However if it were to halt for a fixed time every hour, its average speed comes out to be 50 km/h. For how much time does the train halt for every hour?

A. 6 min

B. 10 min

C. 12 min

D. none of these



10. In a reaching the Purnagiri a man took half as long again to climb the second third as he did to climb the first third and a quarter as long again for the last third as for the second third. He took altogether 5 hrs 50 minutes.

Find the time he spent on the first third of the journey?

A. 72 min

B. 80 min

C. 81 min

D. 88 min

11. Mallah can row 40 km upstream and 55 km downstream in 13 h and 30 km upstream and 44 km downstream in 10 hours. What is the speed of Mallah in still water?

A. 6 km/h

B. 12 km/h

C. 3 km/h

D. 8 km/h



12. In a 600 m race Prabhat has a start of 200 m and the ratio of speeds of Prabhat and Vikram is 4 : 5, then the distance by which Prabhat wins by:

A. 100 m

B. 80 m

C. 120 m

D. None of these

Starting Point		600 m		End Point
Vikram	200 m	Prabhat	400 m	

We, know distance travelled is proportional to speed

Given, ratio of their speed P:V::4:5

So, when Prabhat is travel 400m, vikram will travel 500 m

Hence, Prabhat wins by 100 m.

13. The difference between downstream speed and upstream speed is 3 km/h and total time taken during upstream and downstream is 3 hours. What is the downstream speed, if the downstream and upstream distance is 3 km each?

- A. 2.5 km/h B. 4.33 km/h C. 4 km/h D. 3.3 km/h**
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14. A train overtakes two girls who are walking in the opposite direction in which the train is going at the rate of 3 km/h and 6 km/h and passes them completely in 36 seconds and 30 seconds respectively. The length of the train is:

A. 120 m

B. 150 m

C. 125 m

D. none of these



15. A man covered half of the distance $3x$ km/h and rest at $5x$ km/h. What is the average speed of the man?

- A. $4x$ km/h B. $3.5x$ km/h C. $3.75x$ km/h D. none of these
- 

16. An athlete takes as much time in running 12 meters as a car takes in covering 24 meters. The ratio of the speed of the athlete and car is.

A. 1:3 B. 1:2 C. 2:3 D. 2:5



17. A farmer traveled a distance of 31 km in 9 hours. He traveled partly on foot at 3 Kmph and partly on bicycle at 5 Kmph. The distance traveled on foot is
- A. 10 B. 21 C. 35 D. 14
- 

18. A runs 20% faster than B and is able to give him a start of 6 meters to end the race in dead heat. What is the length of the race?

A. 35

B. 36

C. 42

D. 46



19. A man complete a journey in 8 hours. He travels first half of the journey at the rate of 24 Kmph and second half at the rate of 32 Kmph. Find the total journey in km.

A. 224

B. 216

C. 232

D. 244



20. A man covered half of the distance $3x$ km/h and rest at $5x$ km/h. What is the average speed of the man?

A. $4x$ km/h

B. $3.5x$ km/h

C. $3.75x$ km/h

D. none of these



21. A train passes through a telegraph post in 9 sec moving with a speed of 54 Kmph. The length of train is.

A. 135 B. 130 C. 125 D. 129



22. Two trains start from a place moving with 45 km/hr and 75 km/hr respectively in the opposite direction. After how many hours they will be 1280 km apart.

A. $10 \frac{1}{2}$

B. $10 \frac{2}{3}$

C. $14 \frac{1}{3}$

D. $12 \frac{2}{3}$



23. A train 110 m long takes three seconds to pass a man. How long is the platform if it passes through it in 15 seconds moving with the same speed.

A. 440 m

B. 430 m

C. 400 m

D. 410 m



24. A train 100 meters long completely crosses a 300 meters long bridge in 45 seconds. What is the speed of the train.

A. 32 km/h

B. 40 km/h

C. 36 km/h

D. 48 km/h



25. A train sets off at 2 *pm* at the speed of 70 *km/hr*. Another train starts at 3:30 *pm* in the same direction at the rate of 85 *km/hr*. At what time the trains will meet.

A. 9:30 pm

B. 8:30 pm

C. 10:30 pm

D. 10:45 pm



26. A train 120 m in length passes a pole in 12 sec and another train of same length traveling in opposite direction in 20 sec. Find the speed of the second train.

A. 6.2 km/h

B. 6.4 km/h

C. 7.2 km/h

D. 7.4 km/h



27. Two stations A and B are 110 km apart on a straight line. One train starts from A at 7 am and travels towards B at 20 Kmph. Another train starts from B at 8 am and travel towards A at a speed of 25 Kmph. At what time will they meet?

A. 10 am

B. 10 pm

C. 11 am

D. 11 pm



28. A railway passenger covers the poles on the line as he passes them. If they are 50 m apart and the train is moving at 48 Kmph. How many poles will he pass per minute?

A. 21 B. 19 C. 15 D. 17

29. Two stations A and B are 110 km apart on a straight line. One train starts at 8 am from A and travel towards B at 40 Kmph. Another train starts from B at 10 am and travel towards A at 50 Kmph. At what time will they meet?

A. 10:20 am B. 11:20 am C. 10:20 pm D. 11:20 pm



30. A train running at 25 Kmph takes 18 sec to pass a platform. Next it takes 12 sec to pass a man walking at 5 Kmph in the opposite direction. Find the difference in the length of the train and that of the platform?

A. 25 m

B. 30 m

C. 35 m

D. 20 m



31. The speed of boat upstream is 50 km/hr and downstream is 30 km/hr. Find the velocity of stream.

A. 5 km/h

B. 10 km/h

C. 15 km/h

D. 20 km/h

**32. The speed of a boat in still water is 18 Kmph and the rate of current is 14 Kmph.
Find the distance traveled in downstream in 15 minutes.**

A. 8 km

B. 7 km

C. 9 km

D. 8.5 km

33. Speed of a boat in still water is 6 Kmph and the speed of the stream is 1.5 Kmph. A man rows to a place at a distance 22.5 km and comes back to the starting point. The total time taken by him is.

A. 5 hours

B. 6 hours

C. 8 hours

D. 10 hours



34. The current of a stream flows at 1 Kmph. A motor boat goes 35 km upstream and back to the starting point in 12 hours. The speed of the motor boat in still water is?

A. 8 km/h B. 5 km/h C. 7 km/h D. 6 km/h



35. A man can swim 5 Kmph in still water. If the velocity of the stream is 3 Kmph, the time taken by him to swim to a place 16 km upstream and back is?

A. 11 hrs

B. 10.5 hrs

C. 10 hrs

D. 12.5 hrs



36. The speed of a boat in still water is 4 Kmph and the upstream speed of the stream is 1.5Kmph. Then the speed of the boat going up stream is?

A. 2 km/h

B. 1 km/h

C. 5 km/h

D. 4 km/h



37. A boat covers 24 km upstream and 36 km downstream in 6 hours. If it covers 36 km upstream and 24 km downstream in 6.5 hours. Find the velocity of the current in both the directions.

A. 8 km/h

B. 14 km/h

C. 9 km/h

D. 12 km/h



38. A man can row 6 Kmph in still water. It takes him twice as long to row up as to row down in river. Find the rate of stream?

A. 2 km/h

B. 3 km/h

C. 4 km/h

D. 1 km/h



39. A man rows to a place 48 km distance and back in 14 hours. He finds that he can row 4 km with the stream in the same time as 3 km against the stream. The rate of the stream is?

A. 6 km/h

B. 8 km/h

C. 1 km/h

D. 5 km/h



40. A man rows at a speed of 8 Kmph in still water to a certain distance upstream and back to the starting point in a river which flows at 4 Kmph. Find his average speed for the whole journey?

A. 12 km/h

B. 18 km/h

C. 24 km/h

D. 6 km/h